

Credit - Drishti Ias

Nepal Flash Floods and Vulnerability of the Himalayan Region



For Prelims: Flash floods , Western Disturbance , Kosi, Gandak , Permafrost thaw , Glacial Lake Outburst Floods , Sendai Framework

For Mains: Himalayan geomorphology and tectonic instability, Climate change and cryospheric changes in the Himalayas, Environmental governance in ecologically fragile regions

Source: IE

Why in News?

Catastrophic **flash floods** in Nepal have killed over 150 people, left hundreds missing and damaged critical infrastructure. Scientists suspect an **ice-rock avalanche**, possibly interacting with an anomalous **Western Disturbance** , triggered the sudden flood.

- Nepal's rivers such as the **Bhote Koshi, Gandak and Karnali** originate in or pass through the Himalayan region before entering India. Therefore, floods in Nepal can also create downstream risks for India.
- The event highlights the vulnerability of the **Himalayan region** to compound hazards such as glacier instability, landslides and flash floods.

Summary

- **Nepal's catastrophic flash floods** , suspected to involve an **ice-rock avalanche and anomalous weather interactions** , highlight the increasing exposure of the Himalayan region to **compound hazards** such as glacier instability, landslides and flash floods. These risks are amplified by **tectonic fragility, climate change, unplanned infrastructure and regulatory gaps** .
- The experience underscores the need for **risk-sensitive Himalayan development** through ecological carrying-capacity assessments, cumulative impact evaluation, springshed protection, nature-based slope stabilisation and **multi-hazard early-warning systems** , rather than treating disaster resilience as an afterthought.

What are the Factors Driving the Growing Vulnerability of the Himalayan Region to Frequent Disasters?

Geological & Climatic Drivers (Natural Fragility)

- **Tectonic Stress:** The Himalayas are young, continuously uplifting fold mountains traversed by highly active thrust faults (**Main Central Thrust (MCT)**, **Main Boundary Thrust (MBT)** and **Himalayan Frontal Thrust (HFT)**).
 - Most of the Indian Himalayan Region falls in **Seismic Zone VI**. Furthermore, historical "seismic gaps" (segments without major earthquakes for over a century) store immense unreleased energy, and fractured rocks make the region highly susceptible to mass wasting.
 - The devastating **2015 Gorkha Earthquake in Nepal** triggered thousands of secondary **landslides** across the region due to the fragile, fractured nature of the high-altitude rock faces.
- **Climate Amplification: Elevation-Dependent Warming (EDW)** is accelerating **permafrost thaw** (removing the "ice-cement" holding rocks together) and expanding moraine-dammed glacial lakes, severely increasing the risk of **Glacial Lake Outburst Floods (GLOFs)** and massive ice-rock avalanches.
 - The **2021 Chamoli disaster in Uttarakhand** was triggered by a massive hanging glacier and rock face collapsing due to permafrost thaw, while the **2023 South Lhonak Lake disaster in Sikkim** was a catastrophic GLOF where a moraine-dam breached, washing away downstream towns.
- **Hydrological Extremes:** Shifting monsoons cause **localized cloudbursts** . Combined with steep river gradients, this generates violent, high-velocity flash floods and debris flows that grade riverbeds and shift channels.
 - The **2013 Kedarnath disaster** was heavily amplified by a localized cloudburst and the breaching of Chorabari Lake.

Anthropogenic Drivers (Human Amplification)

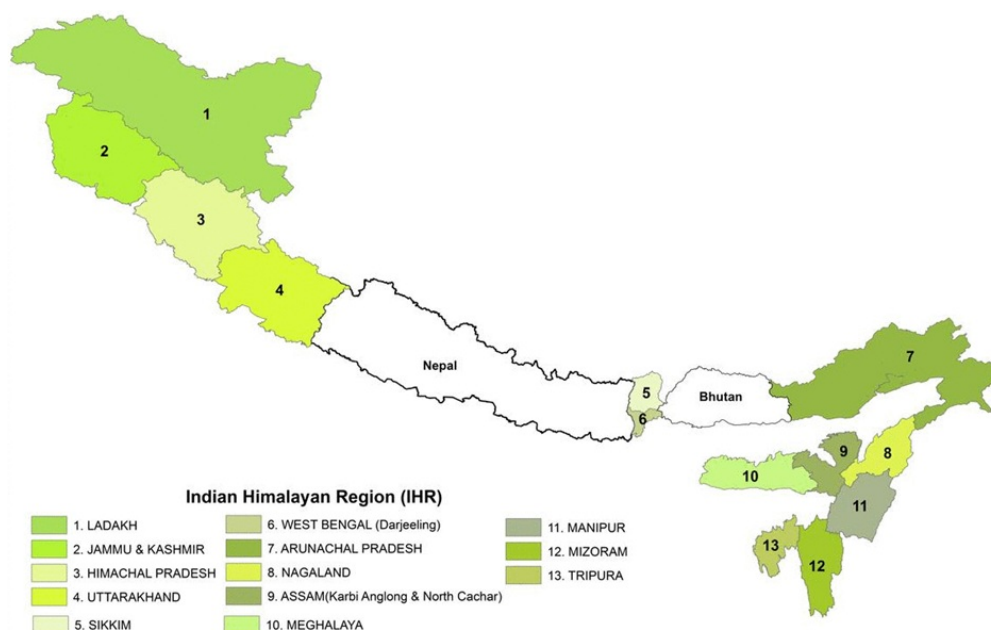
- **Reckless Infrastructure:** Vertical toe-cutting of slopes without retaining walls destabilizes the mountainside. Simultaneously, **mass tourism** has driven heavy concrete urbanization on top of

unconsolidated debris and active floodplains.

- The **Char Dham Pariyojana highway widening** has triggered recurring, massive landslides (like at Lambagarh) due to unscientific vertical hill-cutting. The **Supreme Court-appointed High Powered Committee (HPC)** noted that the **Char Dham Pariyojana** could destabilise the fragile Himalayan ecosystem.
- **Hydrogeological Destruction:** Subterranean tunneling and blasting for run-of-the-river hydropower projects puncture deep aquifers. This drains perched water tables, altering pore-water pressure and causing severe ground subsidence (e.g., the Joshimath crisis).
- **Debris Mismanagement:** Excavated muck from tunneling and road widening is routinely dumped directly into riverbeds, transforming high-discharge floods into lethal debris torrents that destroy downstream dams (e.g., Teesta-III).

Governance & Regulatory Failures

- **Piecemeal Clearances:** The dilution and weakening of the **Environmental Impact Assessments (EIAs) framework**, along with project-wise assessments, can overlook the **cumulative environmental impacts** of multiple infrastructure projects in fragile Himalayan river basins.
 - Individual projects are assessed separately, while **basin-wide Strategic Environmental Assessments (SEAs)** remain limited.
 - This can increase **slope instability, habitat loss and disaster risks** by failing to account for the combined impact of roads, hydropower and urbanisation.
- **Non-Enforcement on Floodplains:** Despite the **2016 Ganga Authorities Order**, states fail to restrict illegal multi-story construction on active river terraces and **"No-Development Zones."**
- **Dilution of Protections:** Relaxations to the **Bhagirathi Eco-Sensitive Zone and exemptions under the Forest Conservation (Amendment) Act 2023** allow large-scale strategic and linear projects to bypass critical ecological scrutiny and slope-stability assessments .



Key Committees Recommendations for Disaster Management in India

- **Mishra Committee (1976):** It was set up to investigate the sinking of Joshimath. It recommended restricting construction in **unstable and landslide-prone zones** , pending detailed site investigations. It also advised against indiscriminate **tree cutting and boulder removal** for construction and road works.
- **J.C. Pant Committee (1999):** Classified **31 disasters into five categories** and recommended:
 - Integrating disaster management into the **Seventh Schedule** and enacting national and State disaster laws.
 - Stronger enforcement of **building codes and safety standards** .
 - Establishing dedicated **institutional mechanisms** for disaster management and coordination.
 - Creating specialised institutions for **disaster training and capacity building** .
 - Strengthening **disaster financing** for response and prevention.
 - Promoting a **culture of preparedness** through risk assessment, training and Standard Operating Procedures (SOPs).

What Measures can be Adopted to Mitigate Disaster Risks in the Indian Himalayan Region?

- **Enforcing Ecological Carrying Capacity:** Legally mandate **Strategic Environmental Assessments (SEA)** and landscape-level cumulative impact assessments before approving major linear or hydel infrastructure.
 - Strict implementation of **River Regulation Zones (RRZ)** to prohibit permanent construction on active river floodplains.
- **Springshed Management & Water Security:** Scale up the **Eight-Step Springshed Management Methodology** (endorsed by **NITI Aayog**) using hydrogeological mapping to demarcate spring recharge areas and prevent artificial tunneling in recharge zones.
- **Bio-Engineering for Slope Stabilization:** Transition from heavy, rigid concrete retaining walls to **nature-based slope stabilization** using deep-rooting grasses (e.g., Vetiver), hydroseeding, geo-textiles, and contoured vegetative terraces.
- **Modernizing Multi-Hazard Early Warning Systems (MHEWS):** Integrate **InSAR (Interferometric Synthetic Aperture Radar)** , satellite telemetry, high-altitude Automated Weather Stations (AWS), and Doppler radars to extend real-time warning windows for downstream communities.
- **Sustainable Mountain Tourism Governance:** Shift from high-density, unregulated tourism to **regulated, high-value low-impact eco-tourism** , enforced through pre-registration caps, mandatory waste buy-back mechanisms, and tourist taxation earmarked for local conservation.
- **Mainstreaming Climate and Disaster Resilience in Development:** Developmental projects in the IHR must integrate disaster risk reduction as per the **Sendai Framework** and climate adaptation to prevent increasing hazard exposure, especially in infrastructure and urban planning.

Conclusion

The recurring Himalayan disasters expose a **development-governance mismatch**, where fragile geology and climate change are being compounded by infrastructure that ignores ecological limits. The challenge is not to stop development, but to align it with the Himalayas' **carrying capacity, cumulative risks and geological, hydrological and cryospheric limits**, ensuring disaster resilience becomes an integral part of development planning.

Drishti Mains Question:

The increasing frequency of Himalayan disasters reflects not merely climate change, but the interaction of geological fragility with developmental pressures. Examine.

Frequently Asked Questions (FAQs)

1. What makes the Indian Himalayan Region highly vulnerable to disasters?

Its **active tectonics, fractured geology, steep slopes, glacier instability, extreme rainfall and increasing anthropogenic pressure** create high multi-hazard vulnerability.

2. What is a Glacial Lake Outburst Flood (GLOF)?

A GLOF is the **sudden release of water from a glacial lake**, often following failure of an ice or moraine dam due to avalanches, earthquakes or excessive inflow.

3. What is the significance of the Sendai Framework for Himalayan disaster management?

The **Sendai Framework 2015-2030** emphasises understanding disaster risk, strengthening governance, investing in resilience and improving preparedness for effective response and recovery.

4. What did the Mishra Committee (1976) recommend regarding Himalayan development?

It recommended restricting construction in **unstable and landslide-prone areas** and avoiding indiscriminate tree cutting and boulder removal during construction and road works.

5. How can disaster risks in the Himalayan region be reduced?

Key measures include **carrying-capacity-based planning, cumulative impact assessment, springshed management, bio-engineered slope stabilisation, multi-hazard early-warning systems and climate-resilient infrastructure**.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/giUWJB0mypo](https://www.youtube.com/embed/giUWJB0mypo)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims :

Q. Consider the following pairs: (2020)

Peak - Mountains

1. Namcha Barwa - Garhwal Himalaya
2. Nanda Devi - Kumaon Himalaya
3. Nokrek Sikkim - Himalaya

Which of the pairs given above is/are correctly matched?

(a) 1 and 2

(b) 2 only

(c) 1 and 3

(d) 3 only

Ans: (b)

Q. If you travel through the Himalayas, you are likely to see which of the following plants are naturally growing there? (2014)

1. Oak
2. Rhododendron
3. Sandalwood

Select the correct answer using the code given below :

(a) 1 and 2 only

(b) 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Ans: (a)

Q. When you travel in Himalayas, you will see the following: (2012)

1. Deep gorges
2. U-turn river courses
3. Parallel mountain ranges
4. Steep gradients causing landsliding

Which of the above can be said to be the evidence for Himalayas being young fold mountains?

(a) 1 and 2 only

(b) 1, 2 and 4 only

(c) 3 and 4 only

(d) 1, 2, 3 and 4

Ans: (d)

Mains:

Q1. Differentiate the causes of landslides in the Himalayan region and Western Ghats. **(2021)**

Q2 . How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? **(2020)**

Q3. “The Himalayas are highly prone to landslides.” Discuss the causes and suggest suitable measures of mitigation. **(2016)**

Transparency Concerns over Collegium System of Judicial Appointments



For Prelims: Collegium System , Chief Justice of India , National Judicial Appointments Commission , Basic Structure

For Mains: Collegium System and Judicial Appointments, Judicial Independence and Separation of Powers, Judicial Accountability and Transparency

Source: TH

Why in News?

The debate over **transparency in judicial appointments** has resurfaced after **the Supreme Court** observed that greater openness in the **collegium process** could strengthen **public confidence** and ensure that **merit remains the governing principle** .

- The issue is significant because the collegium system, while intended to protect **judicial independence from political interference** , has often been criticised for functioning with limited public scrutiny.

Summary

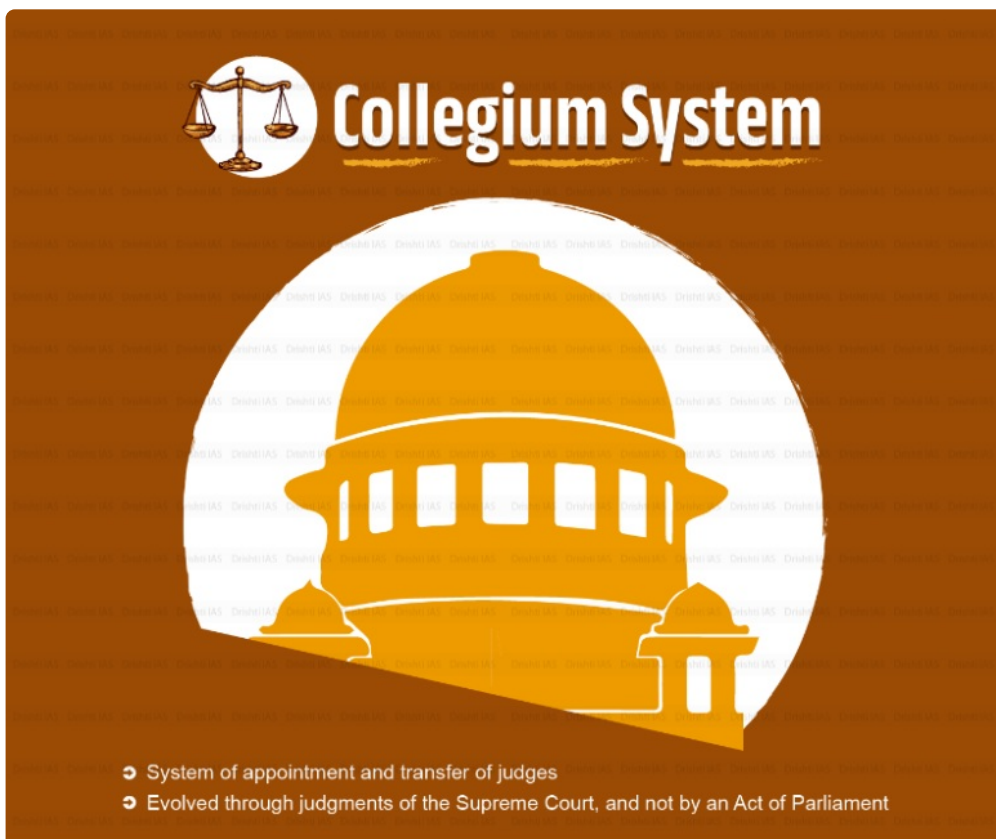
- The **Collegium system** , evolved through the Three Judges Cases, protects judicial independence but faces criticism over **opacity, lack of objective criteria, limited disclosure of reasons and allegations of nepotism** .
- The way forward is to balance **independence with accountability** through **pre-notified vacancies, objective evaluation criteria, a transparent MoP, institutional screening and reasoned decisions while protecting sensitive information** .

What is the Collegium System of Judicial Appointments?

- **Nature of the System:** The Collegium system is a judicially created mechanism—not mentioned in the text of the Constitution of India—responsible for the appointment and transfer of judges to the Supreme Court and High Courts.
- **Constitutional Backdrop:**
 - **Article 124(2):** Appointments to the Supreme Court are made by the President after "consultation"

with judges of the Supreme Court and High Courts.

- **Article 217(1):** Appointments to High Courts are made by the President after consultation with the **Chief Justice of India (CJI)**, the State Governor, and the Chief Justice of the High Court.
- **Evolution through the "Three Judges Cases":**
 - **First Judges Case (S.P. Gupta v. Union of India, 1981):** The Supreme Court ruled that "consultation" does not mean "concurrence," giving primacy to the Union Executive in judicial selections.
 - **Second Judges Case (Supreme Court Advocates-on-Record Association v. Union of India, 1993):** The Court reversed its earlier decision, interpreting "consultation" as "concurrence" and creating the Collegium (CJI + two senior-most judges).
 - **Third Judges Case (Special Reference 1 of 1998):** Expanded the Collegium to its present structure:
 - The SC collegium is headed by the CJI (Chief Justice of India) and comprises four other senior most judges of the court.
 - A High Court collegium is led by the incumbent Chief Justice and two other senior most judges of that court.
 - **Fourth Judges Case (2015):** The Supreme Court struck down the 99th Constitutional Amendment Act and the **National Judicial Appointments Commission (NJAC) Act, 2014** , upholding the Collegium to protect judicial independence as part of the **Basic Structure**.



Collegium System

- System of appointment and transfer of judges
- Evolved through judgments of the Supreme Court, and not by an Act of Parliament

Constitutional Provisions Related to Appointment of Judges

- **Articles 124 (2) and 217-** Appointment of judges to the Supreme Court and High Courts
 - President makes appointments after consulting with "such judges of the Supreme Court and of the High Courts" as s/he may deem necessary.
- But the Constitution **does not lay down any process** for making these appointments.

Evolution of the System

First Judges Case (1981)	Second Judges Case (1993)

- SC held that in the appointment of a judge of the SC or the HC, the word **"consultation"** in Article **124 (2)** and in Article **217** of the Constitution does not mean "concurrence"
- Gave the **executive primacy** over the judiciary in judicial appointments
- SC overruled the First Judges Case
- Gave **birth to the Collegium System (Primacy to the Judiciary)**
- Collegium included the Chief Justice of India and the **2** most senior judges of the SC

Third Judges Case (1998)

- SC expanded the Collegium to include the CJI and the **4** most-senior judges of the court after the CJI

Current Structure	Criticism
 Supreme Court Collegium: CJI and the 4 senior-most judges of the SC  High Court Collegium: CJI and 2 senior most judges of the SC	• Opaqueness • Scope for Nepotism • Exclusion of Executive • No Predetermined Procedure of Appointment

National Judicial Appointments Commission (NJAC)

- It was an attempt to replace the Collegium System. It prescribed the procedure to be followed by the Commission to appoint judges
- NJAC was established by the **99th Constitutional Amendment Act, 2014**
- But the **NJAC Act was termed unconstitutional** and was struck down, citing it as having affected the independence of the judiciary

 **Drishhti IAS**



Why is the Collegium Facing Criticism Over Transparency?

- **Rollback of Reasoned Resolutions and Member Details:** In October 2017, the Supreme Court Collegium resolved to publish resolutions on its website detailing brief reasons for recommendations.
 - In **November 2024**, resolutions stopped publishing both the reasoning behind decisions and the names of the Collegium members who participated.
 - In November 2025, then-CJI confirmed the Collegium resolved to stop publishing detailed reasons to avoid potential damage to the career prospects of unselected candidates (**a justification critics argue would never be accepted for other public offices**).
- **Absence of an Objective Evaluation:** There is no codified framework explaining how candidates are assessed on legal competence, integrity, judgment quality, disposal rates, or social diversity.
 - Upcoming judicial vacancies are not formally advertised, preventing qualified lawyers from applying directly.
- **Disadvantage to First-Generation Lawyers:** A closed-door nomination process structurally favors established networks and disadvantages first-generation advocates who lack institutional connections.
- **The "Uncle Judges" Controversy and Kinship Perceptions:** Opacity fuels recurring allegations that

relatives of sitting and retired judges receive preferential consideration.

- **Allahabad High Court (2018):** The Union Government flagged 11 of 33 recommended names due to familial ties with sitting or retired judges.
- **Supreme Court Composition (2025):** An assessment found that nearly 30% of sitting Supreme Court judges (approximately 10 of 33) had familial links to former judges.
- In January 2025, the Collegium discussed barring judges' relatives before agreeing in principle to subject such candidates to stricter scrutiny.
- **Lack of Diversity Metrics and Audits:** The current Memorandum of Procedure (MoP) for appointing judges does not require disclosure of candidates' demographic data. Without regular audits, monitoring progress or ensuring accountability for inclusivity remains difficult.
- **Internal Jurisprudential Contradictions:**
 - **MediaOne Judgment (2023):** The Supreme Court declared sealed-cover procedures and uncommunicated reasons **"antithetical to a transparent and accountable system,"** yet administrative appointments follow a closed-door model.
 - **Secretary, State of Karnataka v. Umadevi (2006):** The Court mandated that public appointments must follow open and non-arbitrary procedures under Articles 14 and 16.
 - **Subhash Chandra Agarwal Judgment (2019):** While the CJI's office was brought under the **Right to Information (RTI) Act, 2005** substantive Collegium deliberations remain exempt from disclosure.
- **Erosion of Institutional Legitimacy in the Digital Era:** Social media has made judicial functioning subject to instant and decentralised public scrutiny. This makes transparency and clear reasoning increasingly important for maintaining public trust and addressing criticism constructively.

Arguments for Confidentiality (In Favor of Secrecy)

- **Protecting Professional Reputation:** Disclosing reasons for rejecting an advocate or judicial officer could damage their ongoing practice, standing, or judicial morale.
- **Ensuring Candour Among Collegium Members:** Closed-door deliberations allow judges to assess candidates' integrity, judicial temperament, and competence without fear of public fallout.
- **Safeguarding Judicial Independence:** Preventing executive or public interference helps preserve the separation of powers under Article 50.

Global Practices in Judicial Appointments

Country	Mechanism	Transparency & Accountability Features
United Kingdom	Judicial Appointments Commission (JAC)	An independent statutory body; public vacancy advertisements; standardized competency tests; structured interviews; includes lay members.
South Africa	Judicial Service Commission (JSC)	A multi-stakeholder constitutional body; publicly invites nominations and conducts televised public interviews of shortlisted candidates.
United States	Presidential Nomination & Senate Confirmation	Open executive nomination followed by public vetting and confirmation hearings before the Senate Judiciary Committee.

What Structural Reforms Can Modernize the Collegium System?

- **Pre-Notifying Anticipated Vacancies:** Announce vacancies systematically to give all eligible candidates sufficient notice to apply or be recommended.
- **Establishing an Objective Evaluation Matrix:** Lay down measurable criteria for assessing domain expertise, reported judgments, disposal rates, integrity, and social diversity.
- **Institutional Research Secretariat:** Create a permanent screening secretariat to conduct verifiable background checks and data compilation before Collegium meetings.
- **Publishing Sanitized Reasoned Decisions:** Provide summary explanations for elevations and rejections while keeping sensitive intelligence or personal data confidential.
- **Revising the Memorandum of Procedure (MoP):** Finalize a clear, rule-based MoP that establishes binding timelines for both the judiciary and the Union Executive.

Conclusion

Judicial independence and transparency are **complementary, not contradictory**. While independence protects judges from political and external interference, transparency promotes fairness, accountability and public confidence. Therefore, the collegium need not disclose sensitive information, but should ensure **greater openness in its procedures, criteria and broad reasoning** behind appointments.

Drishti Mains Question:

Judicial independence cannot become a justification for institutional opacity. Discuss how the Collegium system can be made more transparent without compromising judicial independence.

Frequently Asked Questions (FAQs)

1. What is the Collegium system?

It is a **judicially evolved mechanism** for appointing and transferring judges of the Supreme Court and High Courts, developed through the **Three Judges Cases** .

2. What are Articles 124(2) and 217(1) related to?

Article 124(2) deals with appointment of Supreme Court judges, while **Article 217(1)** deals with appointment of High Court judges by the President after the constitutionally prescribed consultations.

3. What was the significance of the Fourth Judges Case (2015)?

The Supreme Court struck down the **99th Constitutional Amendment and NJAC Act, 2014** , holding that judicial independence forms part of the **Basic Structure of the Constitution** .

4. What are the major transparency concerns surrounding the Collegium?

Key concerns include **limited disclosure of reasons, absence of a codified selection framework, lack of public vacancy notifications and allegations of nepotism** .

5. How can the Collegium system be reformed?

Reforms could include **pre-notifying vacancies, objective evaluation criteria, an institutional research secretariat, reasoned and sanitised decisions, and a clear, time-bound Memorandum of Procedure (MoP)** .

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/we27jqOm9GA](https://www.youtube.com/embed/we27jqOm9GA)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Consider the following statements: (2019)

1. The 44 th Amendment to the Constitution of India introduced an Article placing the election of the Prime Minister beyond judicial review.
2. The Supreme Court of India struck down the 99th Amendment to the Constitution of India as being violative of the independence of judiciary.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans: (b)

Q. What is the provision to safeguard the autonomy of the Supreme Court of India? (2012)

1. While appointing the Supreme Court Judges, the President of India has to consult the Chief Justice of India.
2. The Supreme Court Judges can be removed by the Chief Justice of India only.
3. The salaries of the Judges are charged on the Consolidated Fund of India to which the legislature does not have to vote.
4. All appointments of officers and staff of the Supreme Court of India are made by the Government only after consulting the Chief Justice of India.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 3 and 4 only
- (c) 4 only
- (d) 1, 2, 3 and 4

Ans: (a)

Mains

Q. Critically examine the Supreme Court's judgement on the 'National Judicial Appointments Commission Act, 2014' with reference to the appointment of judges of higher judiciary in India. (2017)

India's Coconut Sector



Source: PIB

Why in News?

Coconut, often referred to as the **“Tree of Life”** , occupies a unique position in India’s agricultural economy, rural livelihoods, and cultural heritage. With policy interventions like the **Coconut Promotion Scheme (Budget 2026-27)** , the sector is moving towards greater **sustainability, value addition, and income enhancement for farmers** .

What is India's Contribution in the Global Coconut Economy?

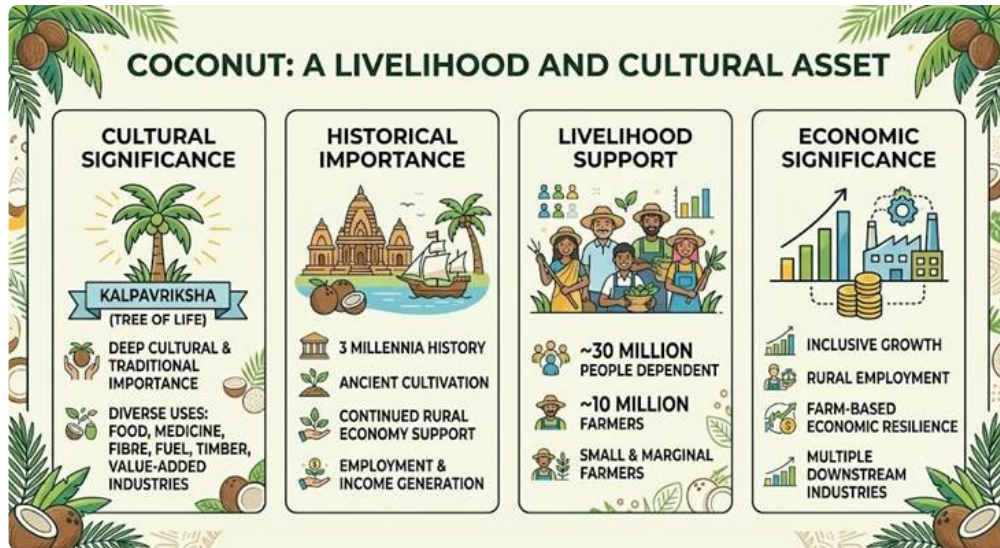
- **Global Position:** India **ranks 1st in coconut production and 3rd in cultivation area** . India contributes **31.24% of global coconut production** and accounts for **17.90% of the global cultivation area** .
- **Production and Productivity:** India produced around **20,301.22 million coconut nuts during 2024-25** from **2.19 million hectares** of cultivation area. The average productivity of coconut in India stood at **9,249 nuts per hectare** during the period.
- **State-wise Contribution:** **Keralam** has the largest area under coconut cultivation. **Tamil Nadu** leads the country in coconut production, while **Andhra Pradesh records the highest productivity** , followed by **West Bengal and Tamil Nadu** .



- **Growing Role in Agricultural Exports:** The coconut sector has become an important contributor to **India’s agricultural exports** and supports the country’s broader export objectives.
 - Coconut product exports reached **₹7,038.35 crore (USD 794.70 million) in 2025-26** , registering a

62% increase compared to ₹4,349.03 crore in the previous year .

- **Transformation of Export Basket:** India's coconut exports have evolved significantly from traditional products such as **coconut oil, copra, desiccated coconut, and coconut shell charcoal** . The sector has diversified into **value-added products** including **virgin coconut oil, coconut milk, coconut water, and activated carbon** .
- **Expansion of Global Markets:** Indian coconut products have expanded beyond traditional markets in the **Middle East** and are now exported to **Europe, the Americas, and Australia** . Major export destinations include the **USA, UAE, Sri Lanka, Germany, Russia, Turkey, Belgium, the UK, and the Netherlands** .



What are the Government Support Measures for the Coconut Sector?

Coconut Development Board (CDB)

- **About:** The **Coconut Development Board (CDB)** was established on **12th January 1981** and functions under the **Ministry of Agriculture and Farmers Welfare** .
 - Headquartered in **Kochi** , the Board works towards the growth and transformation of the coconut sector across **22 States/Union Territories** .
- **Integrated Development:** The CDB is responsible for the **integrated development of coconut cultivation and industry** in India by promoting production, productivity, processing, and market development.
- **Enhancing Production and Sustainability:** The **Coconut Development Board (CDB)** focuses on improving coconut production through the **availability of quality planting material, expansion of cultivation areas, productivity enhancement of existing plantations, and management of major pests and diseases** .
- **Diversification:** The CDB promotes **product diversification, utilisation of coconut by-products, and technology adoption** to strengthen value addition, exports, and income opportunities for coconut farmers.

MSP for Copra

- **Coverage:** **Copra** , the dried kernel of coconut, is included among the **22 mandated agricultural crops covered under the Minimum Support Price (MSP) regime** . The **Union Cabinet** has also

approved an enhanced **MSP for copra** for the **2026 season** to ensure remunerative returns for coconut growers.

- **Types of Copra Produced in India:** India primarily produces two types of copra:
 - **Milling Copra:** It is mainly used for **oil extraction** and is prepared by drying the coconut kernel after removing it from the shell.
 - **Ball Copra (Edible Copra):** It is consumed as a dry fruit and used for religious purposes. It is prepared by drying whole mature coconuts until the kernel hardens inside the shell.
- **Procurement:** The **National Agricultural Cooperative Marketing Federation of India Ltd. (NAFED)** and the **National Cooperative Consumers' Federation (NCCF)** act as the **Central Nodal Agencies** for copra procurement. State-level agencies are also engaged to support procurement operations.

What are the Government Schemes and Initiatives Related to the Coconut Sector?

The Government, primarily through the Coconut Development Board, **implements a range of schemes** to strengthen the coconut ecosystem. These improve access to quality planting material, expand cultivation, enhance productivity, promote technology adoption, and support market development.

Production of Quality Planting Material

- The scheme strengthens the coconut value chain at the foundation stage by ensuring the **availability of quality planting material**. It focuses on establishing **seed production farms, seed gardens, nurseries, and quality certification systems** to improve the supply of healthy coconut seedlings.

Demonstration Cum Seed Production Farms (DSPFs)

- **Establishment of DSPFs:** The **Coconut Development Board (CDB)** establishes **Demonstration cum Seed Production Farms (DSPFs)** across various States to demonstrate **scientific coconut cultivation practices** and produce commercially viable **quality coconut planting materials**.
- **Financial Assistance for New DSPFs:** For newly established DSPFs, financial assistance of **₹30 lakh** is provided from the budget for the **first two years** to support farm establishment and development.
- **Maintenance Support for Existing DSPFs:** From the **third year onwards**, financial assistance of up to **₹1 lakh per hectare per year** is provided for maintenance of DSPFs until yield stabilization, which generally takes **15-20 years**.

Other Initiatives Supporting Quality Planting Material Production

- **Establishment of Nucleus Coconut Seed Garden:** The scheme supports the establishment of coconut seed gardens for producing **high-quality coconut seedlings and improved varieties**.
- **Establishment of Small Coconut Nursery:** The initiative promotes production of quality coconut seedlings through **public and private nurseries**.
- **Accreditation and Rating of Coconut Nurseries:** The scheme ensures the availability of **genuine and high-quality planting material** through nursery accreditation, quality assessment, and certification mechanisms.

- **Assistance for Quality Planting Material Production in Public Sector Nurseries:** The scheme provides financial assistance to eligible public sector nurseries, including **State Departments, State Agricultural Universities (SAUs), Public Sector Undertakings (PSUs), Krishi Vigyan Kendras (KVKs),** and **DSP Farms of the Coconut Development Board** , for strengthening seedling production.

Expansion of the Area Under Coconut Cultivation

- **Objective:** The programme aims to expand coconut cultivation in suitable new areas by providing **financial assistance, incentive-based subsidies, and technical support** for scientific cultivation practices. It also focuses on stabilising existing coconut-growing areas to enhance overall production.
- **Financial Assistance for New Plantations:** Under the scheme, financial assistance of **₹56,000 per hectare** is provided for planting coconut seedlings in new areas. The assistance is released in **two equal annual instalments** .
- **Eligibility under the Scheme:** The subsidy is available for a **maximum area of 2 hectares** and a **minimum area of 0.08 hectares (20 cents)** per beneficiary to encourage wider participation among farmers.
- **Expansion of Coconut Area During 2025-26:** During **2025-26** , major emphasis was placed on expanding coconut cultivation, with around **2,175 hectares of new area** brought under cultivation across several States.
- **States Covered Under Expansion Programme:** The expansion programme covered major coconut-growing and potential States, including **Keralam, Tamil Nadu, Karnataka, Andhra Pradesh, Odisha, West Bengal, Gujarat, Assam, and Arunachal Pradesh** .



Comprehensive Programme for Sustainable Productivity Improvement in Existing Coconut Holdings

- **Objective:** The scheme aims to **improve the production and productivity of existing coconut holdings** through the adoption of **integrated farming practices** , including **coconut-based cropping systems** .
- **Productivity Improvement Through Coconut-Based Cropping System (PICCS):** The **Productivity Improvement Through Coconut-Based Cropping System (PICCS)** is a major component of the programme that promotes sustainable productivity enhancement in coconut plantations.
- **Implementation of PICCS:** The programme is implemented on a **cluster basis** through the **Departments of Agriculture/Horticulture of State Governments** and directly by the **Coconut Development Board (CDB)** .
- **Financial Assistance under PICCS:** The scheme provides **100% subsidy of the cost** , amounting to **₹42,000 per hectare** , with a limit of **2 hectares per beneficiary** . The assistance is released in **two equal annual instalments** .
- **State Government Convergence Support:** Major regions covered under PICCS included **Tamil Nadu, Andhra Pradesh, Keralam, Assam, Goa, and Puducherry** , with **Keralam accounting for a significant share during 2025-26** .
- **Other Components under the Programme:**
 - **Water Resource Creation:** The scheme supports the development of **community tanks, farm**

ponds, and reservoirs to improve irrigation facilities and ensure better water availability in coconut gardens.

- **Organic Farming Promotion:** The programme encourages farmers to adopt **organic farming practices** by providing support for **production, certification, processing, and marketing** of organic products.
- **Farm Mechanisation:** The scheme promotes the adoption of **agricultural machinery and modern technologies** through training, demonstrations, and financial assistance for equipment purchase, post-harvest management, and processing activities.

Replanting and Rejuvenation of Coconut Gardens

- **Objective:** The scheme aims to **enhance coconut production and productivity** by replacing **disease-affected, unproductive, old, and senile coconut palms** with quality seedlings.
- **Rejuvenation of Existing Plantations:** The scheme focuses on **rejuvenating remaining coconut palms** through an **integrated package of practices** to restore productivity and improve the health of coconut gardens.
- **Project-Based Implementation:** The scheme is implemented on a **project-by-project basis** to address **state-specific challenges** and plantation-related issues.
- **Financial Assistance under the Scheme:** Financial assistance of up to **₹54,000 per hectare** is provided over a period of **two years** to support replanting and rejuvenation activities.
- **Major Activities Undertaken:** The key activities under the scheme include:
 - Removal of **diseased and unproductive coconut palms** .
 - Replanting of **quality coconut seedlings** .
 - Rejuvenation and improvement of **existing coconut plantations** through scientific practices.

Complementary Initiatives Strengthening the Coconut Ecosystem

- **Technology Mission on Coconut (TMOC):** The initiative supports the **development and adoption of technologies** for pest and disease management, processing, product diversification, and other technology-based interventions.
- **Kera Suraksha Insurance Scheme:** The scheme provides **accident insurance coverage** to coconut tree climbers, Neera technicians, hybridization workers, and other labourers engaged in the coconut sector.

Coconut Promotion Scheme

- **Objective:** The **Coconut Promotion Scheme** was announced in the **Union Budget 2026-27** to address challenges such as **pest outbreaks, root wilt disease, and aging coconut plantations** , while improving the sustainability of the sector.
- **Enhancing Production and Productivity:** The scheme focuses on **quality planting material production, expansion of coconut cultivation in suitable areas, and replanting/rejuvenation of old and unproductive plantations** to increase productivity.
- **Crop Health Management:** It promotes **integrated crop health management practices** for controlling pests and diseases and improving the resilience of coconut plantations.
- **Value Addition and Financial Support:** The scheme supports **integrated coconut processing units** for value addition and exports, under the broader **₹350 crore allocation for high-value**

agriculture covering coconut, cashew, and cocoa .

- **Long-Term Vision:** The initiative aims to strengthen the **resilience of the coconut sector** , improve farmer livelihoods, and position **India as a global leader in coconut and coir exports by the 2030s** .

Skill Development and Enterprise Creation Initiatives

- **Skill Development Programmes by CDB:** The **Coconut Development Board (CDB)** implements flagship skill development programmes to create a **trained, self-reliant, and enterprise-ready workforce** across activities such as **coconut harvesting, processing, and value addition** .
- **Promotion of Coconut-Based Micro Enterprises:** Through initiatives like Kera Kaushal Kendra, the CDB promotes **coconut-based micro-enterprises, rural entrepreneurship, sustainable livelihoods, and local value addition** .
- **Awareness and Capacity Building:** During **2025-26** , the Coconut Development Board conducted **414 awareness training programmes** covering **CDB schemes, scientific coconut cultivation practices, and coconut value addition techniques** .



Conclusion

India's coconut sector represents the convergence of **agriculture, culture, livelihood security and export potential**. The transformation from a traditional plantation crop to a modern value-added industry demonstrates the importance of integrated policy support. With continued innovation and institutional support, coconut can become a key contributor to **sustainable agriculture, rural prosperity and the vision of Viksit Bharat@2047**.

Drishti Mains Question

India's coconut sector has evolved from a traditional plantation activity to a globally competitive value chain. Discuss.

Frequently Asked Questions (FAQs)

1. Why is coconut called the "Tree of Life"?

Coconut is called the "Tree of Life" because almost every part of the coconut palm is utilized for food, fibre, fuel, medicine, construction and industrial purposes.

2. Which state has the largest area under coconut cultivation in India?

Keralam has the largest area under coconut cultivation in India.

3. Which state has the highest coconut production in India?

Tamil Nadu leads India in coconut production.

4. Which institution is responsible for the development of the coconut sector?

The **Coconut Development Board (CDB)**, established in 1981 under the Ministry of Agriculture and Farmers Welfare, is responsible for integrated development of the coconut sector.

5. What is the Coconut Promotion Scheme?

The **Coconut Promotion Scheme**, announced in Union Budget 2026-27, aims to improve coconut productivity through quality seedlings, plantation rejuvenation, crop health management and processing infrastructure.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/JZ4kl3bmTFs](https://www.youtube.com/embed/JZ4kl3bmTFs)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims :

Q. In the context of India's preparation for Climate -Smart Agriculture, consider the following statements: (2021)

1. The 'Climate-Smart Village' approach in India is a part of a project led by the Climate Change, Agriculture and Food Security (CCAFS), an international research programme.
2. The project of CCAFS is carried out under Consultative Group on International Agricultural Research (CGIAR) headquartered in France.
3. The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in India is one of the CGIAR's research centres.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (d)

Mains :

Q. Explain various types of revolutions, took place in Agriculture after Independence in India. How have these revolutions helped in poverty alleviation and food security in India? **(2017)**

Unified Payments Interface (UPI)



Source: PIB

Why in News?

The **Unified Payments Interface (UPI)**, developed by the **National Payments Corporation of India (NPCI)** , has completed a decade of transforming India's digital payment ecosystem. It has emerged as a key pillar of **India's Digital Public Infrastructure (DPI)** by enabling fast, secure and interoperable payments. **UPI is now operational in 11 foreign countries**, reflecting India's

growing leadership in global digital payment infrastructure.

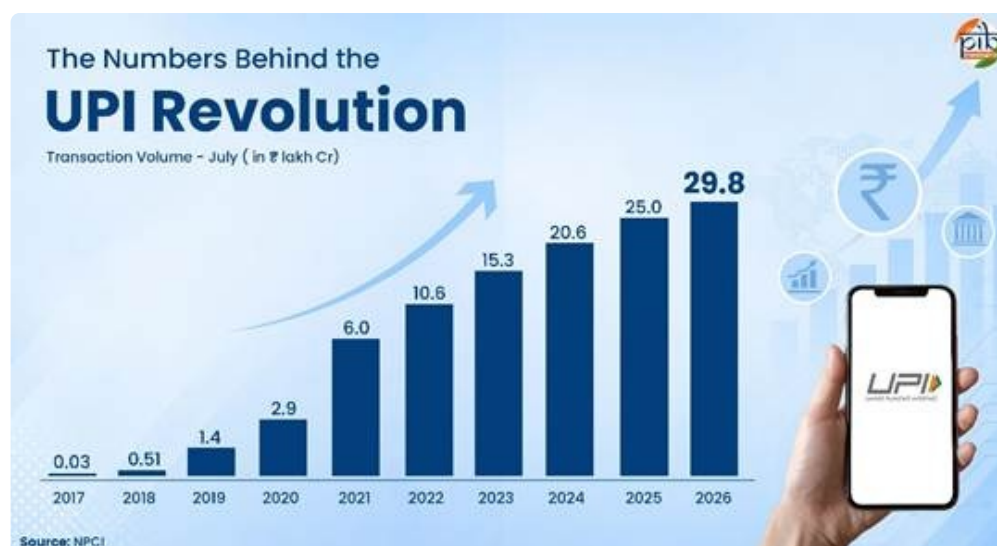
What are the Key Facts About Unified Payments Interface (UPI)?

- **About: Unified Payments Interface (UPI)** is a **real-time payment system** developed by the **NPCI** that enables users to access multiple bank accounts through a single mobile application.
 - It integrates **fund transfers, banking services, and merchant payments** on a single platform and enables **24x7 digital transactions** .
 - UPI ensures secure payments through **two-factor authentication and end-to-end encryption** .
- **Rapid Growth in Transactions:** From **FY 2016-17 to FY 2025-26** , annual UPI transaction volume increased from **2 crore transactions to over 24,162 crore transactions** , registering an approximately **12,000-fold increase** .
- **Growth in Transaction Value:** During the same period, annual transaction value increased from **₹0.07 lakh crore to around ₹314 lakh crore** , reflecting more than a **4,000-fold increase** .
- **Expanding Banking Participation:** As of **July 2026** , the UPI ecosystem includes **741 live banks** , indicating widespread adoption across India's banking sector.
- **High Transaction Scale:** In **July 2026** , UPI processed **2,365.8 crore transactions** with a total transaction value of **₹29.87 lakh crore** .
- **Digital Payment Transformation:** The unprecedented growth of UPI has transformed India's digital payment ecosystem by enabling **fast, interoperable, and accessible financial transactions** for individuals, businesses, and institutions.

Evolution of UPI

- **Launch of UPI (2016):** UPI was launched on a **pilot basis in April 2016 with 21 member banks** and was made available to the public in August 2016 through UPI-enabled banking applications.
 - The **BHIM application** was launched in December 2016 to facilitate direct bank-to-bank transactions using UPI.
- **Expansion of Merchant Payments (2017):** **Dynamic QR codes** were introduced, enabling faster merchant payments by automatically including transaction details and reducing manual entry errors.
- **Introduction of UPI 2.0 (2018):** **UPI 2.0** introduced advanced features such as:
 - **Invoice in the Inbox** for viewing bills before payment authorisation.
 - **Signed Intent and QR** for secure payment approvals.
 - **UPI Mandates** for recurring payments.
 - **Overdraft account linking** for UPI transactions.
- **Wider Adoption (2019):** **SEBI** allowed retail investors to use **UPI as a payment mechanism for IPO applications** .
 - UPI crossed **one billion transactions per month** in October 2019, reflecting rapid adoption.
- **Expansion and Inclusion (2020-2022):** **UPI AutoPay** enabled recurring payments for subscriptions, insurance premiums, SIPs, and EMIs.
 - **UPI 123PAY** brought feature phone users into digital payments through IVR, missed calls, app-based, and proximity sound-based payments.
 - **UPI Lite** enabled faster, low-value, PIN-less transactions.

- Credit cards were integrated with UPI, expanding payment options.
- UPI crossed **5 billion monthly transactions in March 2022** and processed transactions worth over **US\$1 trillion in FY 2021-22**.
- Bhutan marked **UPI's first cross-border deployment**.
- **Credit Integration and Digital Expansion (2023): Credit Line on UPI** was introduced, allowing users to link pre-sanctioned bank credit facilities with UPI.
 - UPI accounted for around **70% of India's digital payment transactions in FY 2023-24**.
- **Enhanced Accessibility and Higher Limits (2024): UPI Circle** enabled primary users to authorise secondary users to make payments within predefined limits.
 - RBI increased the **UPI transaction limit for tax payments from ₹1 lakh to ₹5 lakh per transaction**.
 - The transaction limit for **UPI 123PAY was increased to ₹10,000**.
 - **UPI Lite wallet limit increased to ₹5,000**, with per-transaction limit raised to ₹1,000.
- **Strengthening Security and Inclusion (2025): On-device authentication** enabled payments through smartphone-based fingerprint and face unlock.
 - **Aadhaar-based Face Authentication** simplified UPI PIN onboarding, particularly benefiting first-time users, senior citizens, and individuals without easy access to cards.



UPI's Global Footprint

- **Global Leadership in Real-Time Payments:** As of **2024**, UPI accounted for around **49% of global real-time payment transaction volume**, as highlighted by the **International Monetary Fund (IMF)** in its report *"Growing Retail Digital Payments: The Value of Interoperability"* (June 2025).
- **International Presence:** UPI is operational in 11 foreign countries through UPI acceptance and/or cross-border remittance services. The list includes **Bhutan, Nepal, Singapore, UAE, France, Sri Lanka, Mauritius, Qatar, Cambodia, Greece and Maldives**, enabling seamless cross-border digital payments.
- **Expansion to Cambodia:** In **June 2026**, **NPCI International Payments Limited (NIPL)** partnered with **ACLEDA Bank Plc.** to launch **UPI acceptance in Cambodia**, strengthening India's digital payment connectivity.
- **UPI-Greece Connectivity:** In **June 2026**, UPI went live in **Greece for cross-border remittances**, enabling instant and secure transfers between **Greece and India through Eurobank**.
- **UPI-Maldives Integration:** On **30th July 2026**, cross-border remittances between the **Maldives'**

instant payment system ‘Favara’ and India’s UPI became operational, enhancing bilateral digital financial cooperation.



Conclusion

UPI has transformed India’s payment landscape by making digital transactions **instant, secure, interoperable and inclusive** . From a domestic payment platform launched in 2016, UPI has evolved into a globally recognized digital payment infrastructure. Its rapid adoption demonstrates the potential of India’s **Digital Public Infrastructure model** in promoting financial inclusion and economic transformation.

Drishti Mains Question

Unified Payments Interface (UPI) has emerged as a transformative component of India’s Digital Public Infrastructure. Discuss.

Frequently Asked Questions (FAQs)

1. What is UPI?

UPI is a real-time digital payment system developed by NPCI that enables instant bank-to-bank transactions through mobile applications.

2. Who developed UPI?

UPI was developed by the **National Payments Corporation of India (NPCI)** .

3. When was UPI launched?

UPI was launched on a pilot basis in **April 2016** and publicly rolled out in August 2016.

4. What is UPI 123PAY?

UPI 123PAY enables feature phone users to make digital payments through IVR, missed calls and other methods.

5. How does UPI contribute to financial inclusion?

UPI reduces transaction costs, enables easy digital payments and brings individuals and small businesses into the formal financial system

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/QIWZymIgtT4>

]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. Which one of the following statements about Unified Payments Interface (UPI) and Central Bank Digital Currency (Digital Rupee) is not correct? (2026)

- (a) UPI is a real-time payment system but Digital Rupee is akin to sovereign paper currency.
- (b) In case of UPI, settlement for end users happens instantly as the money gets immediately debited or credited but in case of Digital Rupee, there is no settlement as the wallet balance gets transferred to another wallet.
- (c) UPI transactions are recorded by banks and reflected in bank statements but in case of Digital Rupee, no data is captured in bank statements as transactions are from one wallet to another.
- (d) In both the cases (UPI and Digital Rupee), the liability lies with the users and their respective banks.

Ans: (d)

Q. With reference to digital payments, consider the following statements: (2018)

1. BHIM app allows the user to transfer money to anyone with a UPI-enabled bank account.
2. While a chip-pin debit card has four factors of authentication, BHIM app has only two factors of authentication.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans: (a)

Q. Which of the following is a most likely consequence of implementing the 'Unified Payments Interface (UPI)'? (2017)

- (a) Mobile wallets will not be necessary for online payments.
- (b) Digital currency will totally replace the physical currency in about two decades.
- (c) FDI inflows will drastically increase.
- (d) Direct transfer of subsidies to poor people will become very effective.

Ans: (a)

Mains :

Q. Pradhan Mantri Jan Dhan Yojana (PMJDY) is necessary for bringing unbanked to the institutional finance fold. Do you agree with this for financial inclusion of the poorer section of the Indian society? Give arguments to justify your opinion. (2016)

Smart Border Architecture to Strengthen India's Border Security



Source: TH

The **Ministry of Home Affairs (MHA)** is spearheading a paradigm shift in internal security by developing a **technology-driven, integrated command-and-control system** to establish a **"smart border" architecture** across India's land frontiers.

- **Strategic Focus Area:** The initiative specifically targets regions where physical fencing is not feasible, such as the **3,488-km un-demarcated border with China (Ladakh to Arunachal Pradesh)** and flood-prone **riverine stretches along the Bangladesh border** .
- **Pilot Project:** The initial pilot project will commence along the **Pakistan border** , with a major focus on neutralizing the growing threat of **cross-border drones used for smuggling and reconnaissance** .
- **Four-Pronged Security Strategy:** The architecture integrates the **border guarding forces, local population, State police, and technological solutions** to strengthen monitoring and response capabilities.
- **Technological Integration:** The system creates a common operational picture by linking **Artificial Intelligence (AI), sensors, surveillance cameras, and image-processing tools** to local command centres, which feed live data to field and New Delhi headquarters.
- **Customised Border Solutions:** Recognizing that "**no two borders are the same,**" the deployment will avoid a uniform approach, instead utilizing **terrain-specific technological solutions** for deserts, mountains, and forests, thereby reducing the need for constant physical patrolling.
- **Quadrangular Security Grid:** Following the first-ever **Land Border Districts' Superintendents of Police Conference-2026** , a localized security grid involving border communities and police across **119 border districts (17 States/UTs)** will be finalized within the next 1.5 years.

Read more: [India's Border Security Grid](#)

Khijadiya Bird Sanctuary: Breeding Site for Black-necked Stork



Source: ET

Khijadiya Bird Sanctuary in **Jamnagar, Gujarat** , has emerged as an important breeding stronghold for the **Black-necked Stork** , according to a scientific study published in the *Journal of Threatened Taxa* .

- **Population:** The sanctuary recorded a **peak seasonal population of 21 Black-necked Storks** , including breeding pairs and non-breeding adults.
- **Species:** The **Black-necked Stork (*Ephippiorhynchus asiaticus*)** is one of India's largest wetland birds and has **scattered breeding populations** across the country.
- **Legal Protection:** The species is listed under **Schedule II of the Wild Life (Protection) Act, 1972** , following the **Wild Life (Protection) Amendment Act, 2022** .
 - It is classified as **Near Threatened on the IUCN Red List** , mainly due to **habitat loss and population decline** .
- **Ramsar Site: Khijadiya Bird Sanctuary** , located in **Jamnagar, Gujarat** , was declared a **Wildlife Sanctuary in 1981** and designated a **Wetland of International Importance under the Ramsar Convention in February 2022** . It is recognised for its rich **wetland biodiversity and habitat for resident and migratory birds** .
 - It combines freshwater and saline habitats and lies along the **Central Asian Flyway**, supporting both resident and migratory waterbirds.
- **Conservation Significance:** The findings provide valuable information on the **breeding ecology, nesting behaviour and reproductive success** of the Black-necked Stork, strengthening the scientific basis for its conservation in Gujarat.



Read more: [Protecting the Wetlands](#)

DRDO to Transfer Missile Technology to Indian Defence Industry



Source: TH

Defence Minister has approved the **transfer of DRDO -developed technologies** for all conventional missile systems to Indian defence companies for domestic production, marking a major step towards strengthening **Aatmanirbharta in defence** and expanding **private-sector participation in India's strategic defence manufacturing ecosystem** .

- **About:** The technology transfer will **allow eligible Indian industries to undertake production of DRDO-developed missile systems** after meeting applicable technical qualifications, certifications, and regulatory requirements.
- **Greater Private Sector Participation:** The move will create opportunities for private defence companies, MSMEs, and technology partners to become part of the missile manufacturing supply chain, enabling wider participation in India's defence ecosystem.
- **Objective:** The move aims to accelerate the **transition of missile systems from the development stage to industrial-scale production** , enhance domestic manufacturing capabilities, strengthen the defence industrial base, and increase indigenous value addition.
- **India's Missile Capability Base:** The country possesses a **diverse missile inventory, including the Agni series of ballistic missiles and the BrahMos supersonic cruise missile** , reflecting India's growing indigenous missile development capability.
- **Defence Production:** The initiative forms part of a broader push to deepen private-sector participation in defence manufacturing. **India achieved its highest-ever defence production of ₹1.54 lakh crore in 2024-25** , while defence exports reached a record ₹23,622 crore. The government has set targets to increase defence production to **₹1.75 lakh crore in the current fiscal year and ₹3 lakh crore by 2029** .
- **Significance:** The move will enhance **domestic manufacturing capabilities** , increase **indigenous value addition** , and strengthen India's defence industrial ecosystem by reducing dependence on imported defence equipment.

Read More: [DRDO's Deep Tech Efforts for Defence](#)

Indian Navy to Commission Diving Support Vessel Nipun



Source: PIB

The **Indian Navy** will commission **Nipun** , the **second Diving Support Vessel (DSV)** of the **Nistar-class** , on **31 st August 2026 at Naval Dockyard, Mumbai** . Its induction will significantly enhance India's **deep-sea diving and underwater support capabilities** .

- **About:** Nipun is a specialised platform designed to support a **wide range of deep-sea diving and underwater operations, including diver support, underwater intervention**, salvage missions, and complex maritime tasks.
- **Indigenous Design:** Nipun has been **indigenously designed and constructed by Hindustan Shipyard Limited (HSL), Visakhapatnam** , showcasing India's advancing capabilities in defence shipbuilding under the Aatmanirbhar Bharat initiative.
 - With **nearly 75% indigenous content** , the vessel marks **a significant milestone for India's defence manufacturing ecosystem** and enhances the Indian Navy's capacity to undertake specialised maritime operations.
- **Delivery and Trials:** Constructed as per **Indian Register of Shipping (IRS) classification standards** , Nipun's induction will significantly enhance the Indian Navy's deep-sea diving, underwater intervention, and submarine rescue capabilities.
- **Advanced Diving Capabilities:** The vessel is equipped with **state-of-the-art diving systems, including Deep Sea Saturation Diving capability** and a side diving stage, enabling divers to operate safely in challenging underwater environments.
- **Submarine Rescue Support:** Nipun will serve as a mother ship for **Deep Submergence Rescue Vessels (DSRVs)**, helping evacuate submariners during underwater emergencies and strengthening India's submarine rescue preparedness.
- **Modern Technology and Systems:** The ship features **advanced navigation systems, platform management systems, precise station-keeping capability, remotely operated vehicles (ROVs), and medical support facilities** for sustained underwater missions.
- **Significance:** The name Nipun comes from Sanskrit, meaning “ **skilled, capable, and proficient.**” Its crest, featuring a whale and diving helmet, symbolises endurance, deep-ocean operations, diver safety, and professional excellence in underwater missions.

Read More: [Diving Support Craft A20](#)